

CURRICULUM VITAE



Name: TANISHA RAJ

B.Tech - Metallurgical and Materials Engineering

Roll number: 23MM8013

Gender: Female

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Category: GEN-EWS

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Educational Qualification:

• Graduation

SEM	1	2	3	4	5	6	7	8
SGPA	8.57	8.15	8.77	8.72	8.63	8.52	NA	NA
CGPA	8.57	8.37	8.52	8.57	8.58	8.58	NA	NA

• Higher Secondary & Secondary

Year	Degree/Examination	Institution/Board	CGPA/Percentage
2023	Class XII	Central Board of Secondary Education	94.2
2021	Class X	Central Board of Secondary Education	98.2

Academic Achievements:

- Selected in **SURGE** Programme, **IIT Kanpur** , under **Fully Institute Funded Category (Top 60 nationwide)**
- Selected as a **SheFi Scholar 2026**, a competitive programme focused on **finance, Web3, DeFi, and emerging technologies**.
- Secured **3rd rank** in school at **AISSE-2021 (Class X)**

Skills Summary:

- **Technical Skills :** Python (Pandas, NumPy, Matplotlib, Scikit-learn), SQL (MySQL), Microsoft Excel, Power BI, C/C++, Machine Learning
- **Domain Skills:** Materials Characterization (W-SEM, Optical Microscopy, FTIR, Raman, UV-Visible Spectroscopy), Electrochemical Techniques (OCP, EIS, LPR, Cyclic Polarization), Corrosion Analysis, VersaStudio, OriginPro
- **Soft Skills:** Analytical Thinking, Problem Solving, Attention to Detail, Communication, Leadership, Teamwork, Time Management

Projects/ Internship Details:

- **Summer Research Intern** (May – July 2026)
Summer Undergraduate Research Graduate Excellence (SURGE)
Indian Institute of Technology Kanpur, Uttar Pradesh- [\[Link\]](#)
 - Selected for the SURGE 2026 Research Internship at IIT Kanpur under the guidance of Prof. Kallol Mondal.
 - Conducted electrochemical corrosion studies using **OCP,EIS, LPR and Cyclic Polarization**.
 - Performed material characterization using **W-SEM, FTIR, Raman, and UV–Visible Spectroscopy**.
 - Analyzed corrosion behaviour and **adsorption mechanisms of organic inhibitors for high-chromium steel**.
- **Voltix Insights: Omnichannel Retail Analytics & Profitability Pipeline - [\[GitHub-link\]](#)**
 - **Tools: Python, SQL, Power BI**
 - Engineered an automated **Python ETL pipeline** to clean, deduplicate, and feature-engineer **15K+ retail transactions**, achieving a **100% test pass rate using pytest**.
 - Developed advanced **SQL analytics** using CTEs, window functions (LAG, DENSE_RANK), and subqueries to calculate MoM growth and identify product-level margin leakages.
 - Built an executive **Power BI dashboard** with a custom dark-mode design, dynamic **DAX time-intelligence measures (YoY%, MoM%)**, and interactive tooltips for performance analysis.
 - Identified **sub-15% profit margins** among top-revenue SKUs driven by promotional discounting and delivered recommendations targeting ₹1.2M in annual net-profit recovery.
- **Predicting Yield Strength of Steel Alloys by Regression - [\[github-link\]](#)**
 - **Tools: Python , Numpy, Matplotlib,Scikit-learn**
 - Developed a **regression-based model** to predict **steel properties** and **analyse material behaviour**
 - Applied data preprocessing and optimization techniques for reliable predictions
 - **Visualized property relationships** to **gain insights into strength–ductility trends**

Position Of Responsibility:

- **Convenor** (February 2023- Present)
Indian Society for Technical Education (ISTE), NIT Durgapur
- **Event Management Head** (February 2024- Present)
Indian Institute of Metals (IIM) Student's Chapter NIT Durgapur

Extra-Curricular Activities:

- **PAINTING [UPTO 5TH YEAR]**
CHITRAKALA(SCHOOL OF ART & CRAFT), DURGAPUR
Affiliated to BANGIYA SANGEET PARISHAD & BHARATI KALA KENDRA
- **RECITATION [UPTO 3RD YEAR]**
KATHA O KOBITA DURGAPUR
BANGIYA SANGEET PARISHAD